Hippocampus Volume Quantification for Alzheimer's Progression

Complete Project Report – Udacity AI for Healthcare Nanodegree

3D Medical Imaging · End-to-End Clinical AI System

1. System Architecture

The system follows a three-stage pipeline that mirrors real-world medical imaging AI development: dataset curation, model training, and clinical integration.

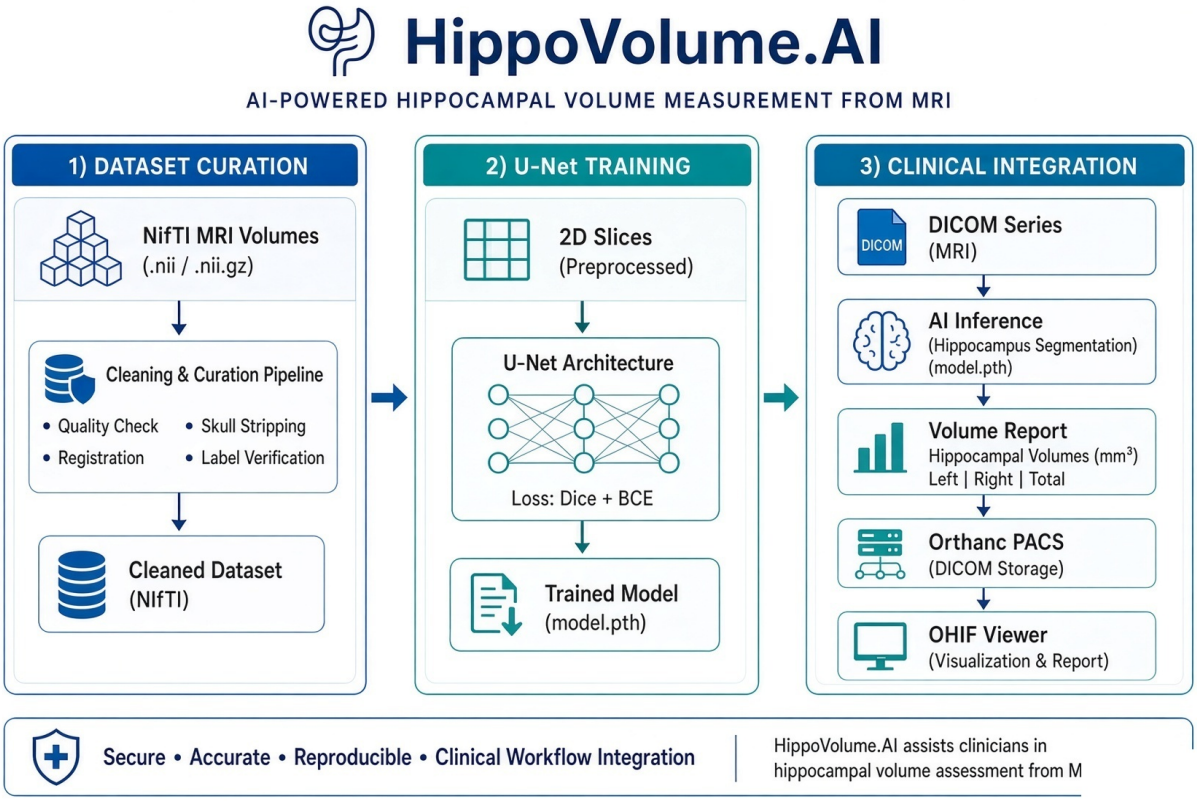


Figure 1. High-level architecture of HippoVolume.AI

2. Section 1 – Curating a Dataset of Brain MRIs

2.1 Dataset Cleaning and Outlier Removal

The Medical Decathlon Hippocampus dataset was inspected. Volumes and corresponding segmentation masks were loaded with NiBabel. Hippocampal volumes (mm^3) were computed as voxel count × voxel volume derived from the NIfTI header (get_zooms). A histogram of volumes revealed extreme outliers (non-brain scans or empty/invalid labels). These were removed using interquartile-range thresholds, yielding a clean set of approximately 260 matched image/label pairs. The cleaned NIfTI collection was saved to section1/out/.

2.2 Medical Metadata Inspection

Header inspection confirmed NIfTI format, bit depth, units (mm), and regular grid spacing. Physical dimensions of the hippocampus were obtained by multiplying the number of labeled voxels by the product of the three voxel spacings. This established the link between array indices and real-world millimeters, a requirement for clinically meaningful volume reporting.

2.3 Slice Visualization

Axial, coronal, and sagittal slices were rendered with Matplotlib, including image-label overlays and maximum-intensity projections. These visualizations validated that pixel data were stored as expected and that the hippocampus appeared in the anticipated location within the cropped volumes.

3. Section 2 – Training a Segmentation CNN

3.1 Training and Validation Pipeline

A 2D U-Net (RecursiveUNet) was trained on axial slices extracted from the curated volumes. All TASK placeholders were completed. The pipeline performed data loading, 60/20/20 split, training with CrossEntropyLoss and Adam, validation, and final testing without errors. The trained weights were saved as model.pth (~119 MB) in section2/out/.

3.2 TensorBoard Monitoring

Scalar loss and sample image grids (input, ground-truth mask, prediction) were logged to TensorBoard. Screenshots of the loss curves and prediction visualizations were captured and included in the output folder, demonstrating that training progress was actively monitored.

3.3 Volumetric Performance Metrics

Dice3D and Jaccard3D were implemented in utils/volume_stats.py. On the held-out test set the model achieved Mean Dice = 0.9004 and Mean Jaccard = 0.8202. Results were written to results.json.

Metric	Value
Mean Dice	0.9004
Mean Jaccard	0.8202

4. Section 3 – Integrating into a Clinical Network

4.1 DICOM Inference Pipeline

inference_dcm.py implements the full clinical pathway: series selection, volume reconstruction from DICOM, U-Net inference (with padding to the training patch size), volume calculation, Secondary Capture report generation, and C-STORE to Orthanc. All TASK items were completed.

4.2 Report Generation and OHIF Visualization

A sample report.dcm containing anterior, posterior, and total hippocampal volumes was generated and successfully displayed in the OHIF viewer alongside the original MRI series. A screenshot of the report in OHIF is included in the submission.

4.3 Validation Plan

A 1–2 page validation plan addresses intended use, ground-truth definition (dual-radiologist consensus via STAPLE), accuracy metrics and success criteria, study population, operating range, and real-world performance estimation. The plan is stored as validation_plan.txt in section3/out/.

5. Clinician-Facing Explanation

An explanatory email was prepared for radiologists. It describes what the algorithm measures, the performance achieved (Dice ≈ 0.90), how the report appears in the PACS, and the appropriate operating conditions (diagnostic-quality axial T2 MRI, slice thickness ≤ 2 mm).

6. Conclusion

The project fulfills all mandatory rubric criteria across the three sections. A clean dataset was produced, a U-Net reached clinically relevant segmentation accuracy, and the model was integrated into a simulated clinical network that delivers a quantitative hippocampal volume report viewable by radiologists. The accompanying validation plan and clinician communication materials support further clinical evaluation.

All code, models, reports, and screenshots are available in the respective section/out folders.